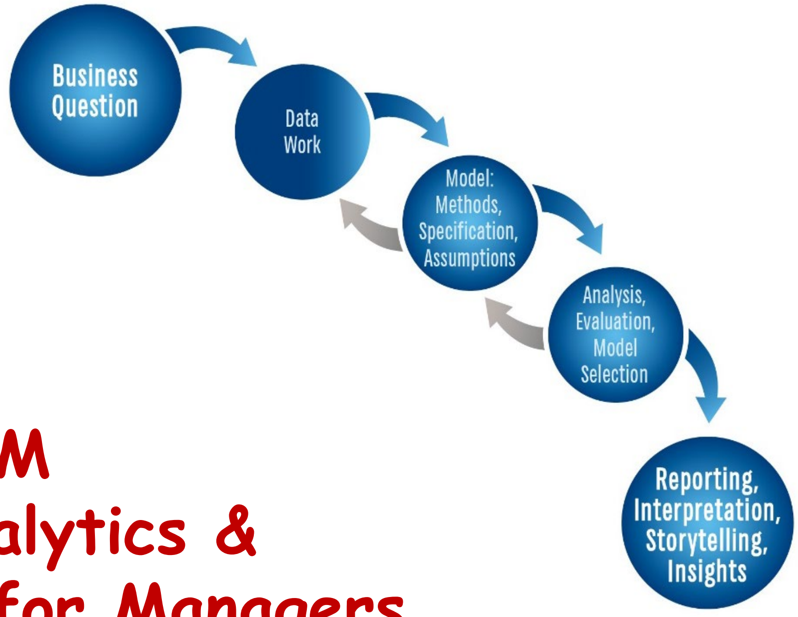

PAML4M

Predictive Analytics & Machine Learning for Managers

9. Classification Models

Use PAML4M_9R_ClassificationModels.Qmd

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Last updated 8/1/2026

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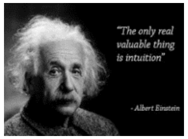
Analytics Modeling Options

	Modeling Method		
	Structured		Visual, Text, Unstructured, etc.
Descriptive	Cluster analysis, correlation, market basket analysis, sample statistics, ANOVA		Bubble charts, network diagrams, natural language processing, clustering dendograms, etc.
Predictive	Association	Decision Tree	Charts
Quantitative Value	Regression	Regression Trees	Regression plots, scatter plots, Tableau diagrams, trend charts, etc.
Classification	Logistic Regression; Other Categorical Regression Models	Classification Trees	Tree maps, interactive diagrams,
Prescriptive	Operations research, decision modeling, optimization, linear programming		Simulations, etc.

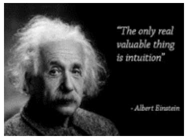
Introduction to Classification Models

YC(x) – Y is not continuous, but categorical



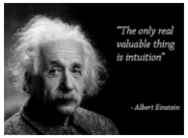
Classification Models: Intuition

- A classification model is one in which the outcome variable is categorical, and the model aims at predicting the category or class of an outcome, based on the likelihood that the outcome falls in that category.
- Examples: is an e-mail message spam or legitimate? is a loan customer likely to default or not? is a given purchase transaction fraudulent or not? how would you predict who will make an A in the class?
- Popular modeling methods for classification include:
  - **Logistic Regression** – Binomial and Multinomial
  - Linear & Quadratic Discriminant Analysis (FYI)
  - **Classification Tree Models** (next week)
  - **Neural Networks** (final week)
  - K Nearest Neighbors (not covered)
  - Support Vector Machines (not covered)



Binomial vs. Multinomial Categorical Outcomes

- Some categories are ordinal – e.g., freshman, sophomore, etc.
- Some are unordered → categorical – e.g., rural, suburban, urban
- We need to transform categories to quantitative values to fit regression models
- If the outcome variable is binomial – i.e., it has two possible outcomes, so it can be easily quantified with dummy variables (e.g., 0 = no loan default; 1 = loan default)
- If the outcome is multinomial – i.e., has k > 2 possible outcomes (e.g., stroke, heart attack, no illness)
  - ➢ It is more difficult to quantify categorical outcomes
  - ➢ We need to transform convert categorical to binary variables



Binomial Classification and MLE

- Maximum Likelihood Estimation (MLE) is a popular estimation method for binomial classification
- For 2 independent observations A and B, the probability of both happening is: $P(A\&B) = P(A) * P(B)$
- So, if an outcome variable can take a value of 0 or 1, the probability of all observations x_i being classified correctly is equal to the product of the probability of each observation being classified correctly, either as 1 or as 0:

$$\boldsymbol{Likelihood(\beta_0, \beta_1)} = \prod_{\boldsymbol{i:\, y_i=1}} \boldsymbol{p(\hat{y}_i | x_i = 1)} \prod_{\boldsymbol{i:\, y_i=0}} \overbrace{\boldsymbol{(1 - p(\hat{y}_i | x_i = 1))}}^{p(\hat{y}_i | x_i = 0)}$$

All positives **All negatives**

- The MLE method computes this likelihood for several values of β_0 and β_1 using algorithms and selects the ones where the likelihood of a correct classification is the largest

Binomial Logistic Models

More on Interpretation

For a model: $\text{Logit}(Y) = \beta_0 + \beta_1(X_1) + \beta_2(X_2) \;\ldots\ldots + \;\varepsilon$

- **Y is a binary variable** (e.g., Non-Smoker = 0; Smoker =1)

  β_0 is the log-odds of $Y = 1$ when $X_1 = X_2 = 0$ (just born, male → smokes)

  e^{β_0} is the odds of $Y = 1$ when $X_1 = X_2 = 0$ (just born, male → smokes)

- **If X_1 is a binary variable** (e.g., Male = 0; Female = 1)

  $\beta_0 + \beta_1$ = log-odds of $Y = 1$ when $X_1 = 1$ and $X_2 = 0$ (just born, female → smokes)

  $e^{(\beta_0+\beta_1)}$ = odds of $Y = 1$ when $X_1 = 1$ and $X_2 = 0$ (just born, female → smokes)

  β_1 = change in log-odds of $Y = 1$ (smokes) when $X_1 = 1$ (female, compared to male)

  e^{β_1} = how much the odds of $Y = 1$ multiply when $X_1 = 1$ (↑ if >1; ↓ if <1)

- **If X_2 is a continuous variable** (e.g., Age)

  β_2 is the increase in log-odds of $Y = 1$ when $X_2 \uparrow 1$ (age increases by 1)

  e^{β_2} is how much the odds of $Y = 1$ multiply when $X_2 \uparrow 1$ (↑ if >1; ↓ if <1)

Interpretation Pointers

- Log Odds (0) → Odds = 1 → Prob = 0.5
  → chance (flip a coin)
- If $\beta = 0$ (i.e., not significant) the predictor doesn't change the odds
- If Log Odds (-) → Odds < 1 → Prob < 0.5
  → (-) β in a logistic regression
  → The log odds go down
  → The odds change by
  less than 1 to 1
- If Log Odds (+) → Odds > 1 → Prob > 0.5
  → (+) β in a logistic regression
  → The log odds go up
  → The odds change by
  more than 1 to 1

Prob	Prob%	Odds	Log Odds
		$p/(1-p)$	$Ln(o)$
0.001	0.1%	0.001	-6.907
0.01	1%	0.010	-4.595
0.1	10%	0.111	-2.197
0.2	20%	0.250	-1.386
0.3	30%	0.429	-0.847
0.4	40%	0.667	-0.405
0.5	50%	1.000	0.000
0.6	60%	1.500	0.405
0.7	70%	2.333	0.847
0.8	80%	4.000	1.386
0.9	90%	9.000	2.197
1.0	100%		

Log Odds	Odds	Prob	Prob%
	$e^{(Log\ Odds)}$	$o/(1+o)$	
-6.907	0.001	0.001	0%
-4.595	0.010	0.01	1%
-2.197	0.111	0.1	10%
-1.386	0.250	0.2	20%
-0.847	0.429	0.3	30%
-0.405	0.667	0.4	40%
0.000	1.000	0.5	50%
0.405	1.500	0.6	60%
0.847	2.333	0.7	70%
1.386	4.000	0.8	80%
2.197	9.000	0.9	90%
		1.0	100%

Binomial Logistic Regression

- {libary} Topic
  `Search keywords`
- Binomial Logistic Regression
  ```
  heart <- read.table("Heart.csv",
  ```

The Confusion Matrix

Is a table that makes it easy to visualize if a model is **"confusing"** or misclassifying categorical predictions

Confusion Matrix: Classification Fit Statistics

$$\text{Error Rate} = \frac{\text{Incorrect}}{\text{Total}} = \frac{FP + FN}{T}$$

$$\text{Accuracy Rate} = \frac{\text{Correct}}{\text{Total}} = \frac{TP + TN}{T}$$

Accuracy predicting positives:

$$\text{Sensitivity} = \frac{TP}{P} = \frac{TP}{TP + FN}$$

Accuracy predicting negatives

$$\text{Specificity} = \frac{TN}{N} = \frac{TN}{TN + FP}$$

Error predicting positives (False Positives)

$$1 - \text{Specificity} = 1 - \frac{TN}{TN + FP} = \frac{FP}{N}$$

Predicted	Actual No (0)	Actual Yes (1)	Total
No (0)	[1,1] TN	[1,2] FN	N_{Pr}
Yes (1)	[2,1] FP	[2,2] TP	P_{Pr}
Total	N	P	T

Classification Threshold

- We generally use the threshold λ = Prob (Y=1) > 0.5 to predict a positive, and the opposite for a negative.
- But if we are more interested in either sensitivity or specificity, we can change this threshold to other values

Binomial Logit: Fit Statistics

- The glm() function in R reports two important fit statistics: (1) Deviance or 2LL and (2) AIC → 2LL + 2n (n = number of predictors)
- Confusion Matrix → A better way to evaluate classification models:

$$\text{Error Rate} = \frac{\text{Incorrect}}{\text{Total}} = \frac{457 + 141}{1{,}250} = 0.48 = 1 - \text{Accuracy Rate}$$

$$\text{Accuracy Rate} = \frac{\text{Correct}}{\text{Total}} = \frac{145 + 507}{1{,}250} = 0.52$$

$$\text{Sensitivity} = \frac{\text{True Positives}}{\text{All Positives}} = \frac{507}{648} = 0.78$$

$$\text{Specificity} = \frac{\text{True Negatives}}{\text{All Negatives}} = \frac{145}{602} = 0.24$$

Predicted	Actual		Total
	Down	Up	
Down	[1,1] 145	[1,2] 141	286
Up	[2,1] 457	[2,2] 507	964
Total	602	648	1,250

$$\text{False Positives} = 1 - \text{Specificity} = \frac{\text{False Positives}}{\text{All Negatives}} = \frac{457}{602} = 0.76$$

- The model above has almost the same error rate as flipping a coin (not good); it's very imprecise about predicting negatives (Down) but it does much better at predicting positives (Up)

Confusion Matrix

- {libary} Topic
  `Search keywords`
- The Confusion Matrix
  `train <- sample(1:nrow(heart),`

Sensitivity, Specificity and ROC

Unbalanced Data

- In some data sets the outcome variable is unbalanced
- For example, when people get tested for Covid, maybe only 5% are positive. This would mean that if we predict that everyone is negative without a model (i.e., null model), our success at predicting negatives (i.e., specificity) would be 95%. But our sensitivity would be 0% (i.e., no positive predictions).
- This means that our model can only improve over the null model if the resulting specificity is > 95%
- And any prediction of positives will improve the sensitivity.
- This is not ideal. One solution is to **sub-sample to balance the data.**
- For example, if the data has 10,000 observations with only 5% (500) positives, draw a **sub-sample** that contains **ALL positives** (i.e., 500) and a **random sub-sample** of an equal number of negatives (500) and use this sub-sample to train your model.
- See: PAML4M_9R_ClassificationModels.Qmd → Unbalanced Outcome Variables

How do we Select λ ?

- Classification models are based on probability
- You can vary the classification threshold $\text{prob} > \lambda$ (e.g., 0.5) and get different results → a tuning parameter
- So, what is the best value of λ? → It's a tradeoff !!
- The specific λ to select within that model to select depends on the analysis goals → True Positives or True Negatives?

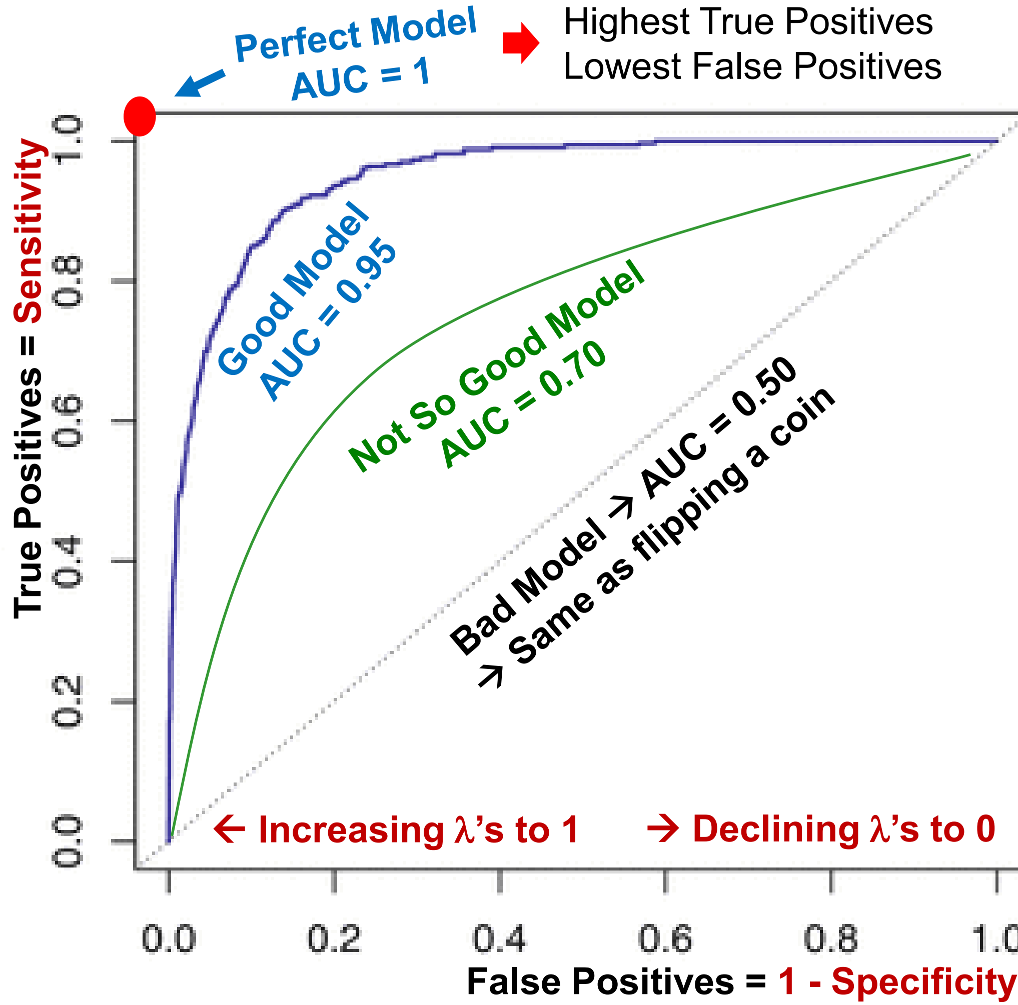
λ, Sensitivity, Specificity and ROC

- As the classification threshold λ changes, sensitivity and specificity move in opposite directions
- $\lambda \uparrow \rightarrow$ Sensitivity $\downarrow$; Specificity $\uparrow$; 1- Specificity $\downarrow$
- You need to try several values of λ from 0 to 1 and compare the resulting Sensitivity and Specificity values
- Naturally, a model that provides large values of both, Sensitivity and Specificity at various values of λ is the best model because it provides more accurate classification, overall
- Receiver Operating Characteristics (ROC) curve helps visualize this tradeoff

ROC (Receiver Operating Characteristics) Curve

- The ROC is a plot constructed by trying several values of λ from 0 to 1 in the model and plotting the resulting Sensitivity (true positives) and 1-Specificity (false positives)

- The larger the "area under the curve" (AUC close to 1) the better the model – i.e., the curve "hugs" the top-left corner.

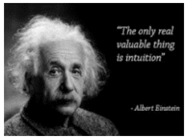


- The dotted 45^0 line represents a model that performs no better than chance with AUC = 0.5

ROC Curve

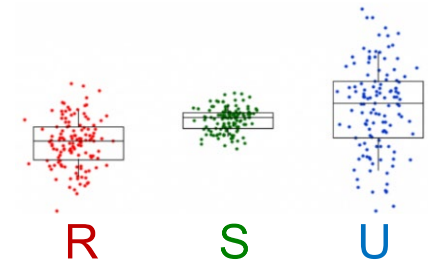
- {libary} Topic
  `Search keywords`
- The Receiver Operating Characteristics (ROC) Curve
  `library(ROCR)`

Multinomial Logistic Models



Multinomial Logistic: Intuition

- In binomial logit the response variable has 2 possible values (e.g., "good mail" and "spam"), so we can easily quantify this variable with 2 binary variables coded 0 (e.g., good mail) or 1 (e.g., spam)
- But if the response variable has more than 2 categories we have a multinomial logistic situation
- For example, what is the likelihood that a person will buy either a rural, suburban, or urban home?



- We could solve this problem fitting 2 binomial models by:
  - ✓ Selecting one home type (e.g., rural) as the "reference" and code all rural home purchases as 0 in both models.
  - ✓ Then fitting two binomial models, one for suburban (1) vs. rural (0) and another for urban (1) vs. rural (0).
- But this is NOT ideal because you would end up with 2 separate regression models with different fit statistics.
- Multinomial logistic regression fits multiple binomial models at once in a single model, with a single set of fit statistics.

Multinomial Logistic Details

- More generally, a response variable with K categories can be modeled as a multinomial model with K-1 response variables and one variable left out (coded as 0) as the **reference** category.
- That is,
  - ➢ The response variable Y has K categories
  - ➢ We select one of them as the reference category k_R
  - ➢ We estimate logistic coefficients for the other $K\text{-}1$ categories
  - ➢ In binomial logistic, a coefficient for X_1 represents the increase in log odds of $Y=1$, relative to $Y=0$, when X_1 increases by 1 unit.
  - ➢ In multinomial logistic, the coefficient for a predictor X_{1k} for the k^{th} category represents the increase in log odds of $Y=k$, relative to $Y=k_R$, when X_1 increases by 1 unit.

Multinomial Logistic Interpretation

Multinomial Logistic Interpretation

For a multinomial logit model:

$$
\begin{aligned}
\text{Logit}(Y) = {} & \beta_{01} + \beta_{11}(X_1) + \beta_{21}(X_2) + \beta_{31}(X_3) + \ldots\ldots \\
& \beta_{02} + \beta_{12}(X_1) + \beta_{22}(X_2) + \beta_{32}(X_3) + \ldots\ldots \\
& \ldots\ldots \\
& \beta_{0K\text{-}1} + \beta_{1K\text{-}1}(X_1) + \beta_{2K\text{-}1}(X_2) + \beta_{3K\text{-}1}(X_3) + \ldots\ldots + \varepsilon
\end{aligned}
$$

- Y has K categories (e.g., Freshman, Sophomore, Junior, Senior withdrawing from school)
- $\text{Logit}(Y)$ is the log odds of the response variable being in category k (e.g., Junior), relative to the reference category k_R (e.g., Freshman)
- β_{ik} is the effect of variable X_i (e.g., GPA) on the log-odds of Y being in category k or $Y = k$ (e.g., Junior), relative to k_R (e.g., Freshman)
- That is, β_{ik} measures how much the log-odds of Y being in category k change when X_i (e.g., GPA) goes up by 1 unit, relative to the log-odds of Y being in the reference category k_R (e.g., Freshman).

Multinomial Logistic Regression

- {libary} Topic
  `Search keywords`
- The Receiver Operating Characteristics (ROC) Curve
  `library(VGAM)`

Multinomial Logit: Fit Statistics

- The `vglm() {VGAM}` function in **R** reports the:
  - **Log-Likelihood**
  - **Deviance** (2LL) → $-2 * Log\ Likelihood$
  - **AIC** → calculated by adding $2 * Number\ of\ Variables$

Confusion Matrix:

$$\text{Accuracy } Rate = \frac{Correct}{Total} = \frac{Diagonal}{T}$$

$$Error\ Rate = \frac{Incorrect}{Total} = \frac{Off - Diagonal}{T}$$

$$Sensitivity_{Class} = \frac{Correct_{Class}}{Total_{Class}} = \frac{T_A}{A}; \frac{T_B}{B}; \frac{T_C}{C}$$

$$Specificity_{Class} = \frac{Correct_{NotClass}}{Total_{NotClass}} =$$

$$\frac{T_B + TC}{B + C}(for\ A); \frac{T_A + TC}{A + C}(for\ B); \frac{T_A + TB}{A + B}(for\ C)$$

Pred	Actual			Total
	A	B	C	
A	T_A	$F_{A/B}$	$F_{A/C}$	P_A
B	$F_{B/A}$	T_B	$F_{B/C}$	P_B
C	$F_{C/A}$	$F_{C/B}$	T_C	P_C
Total	A	B	C	T

Multinomial to Binomial

The **multinomial** confusion matrix is very difficult to interpret. It is easier to analyze it binomially, **category by category** (i.e., A vs. not A, B vs. Not B, C vs. not C, etc.)

Predicted	Actual: Not A	Actual: A	Total
Not A	TN_A	FN_A	N_{Pr}
A	FP_A	TP_A	P_{Pr}
Total	N_A	P_A	T

$$\textit{Error Rate}A = \frac{Incorrect}{Total} = \frac{FP_A + FN_A}{T}$$

$$\textit{Accuracy Rate} = \frac{Correct}{Total} = \frac{TP_A + TN_A}{T}$$

Accuracy predicting positives:

$$\textit{Sensitivity} = \frac{TP_A}{P_A} = \frac{TP_A}{TP_A + FN_A}$$

Accuracy predicting negatives

$$\textit{Specificity} = \frac{TN_A}{N_A} = \frac{TN_A}{TN_A + FP_A}$$

Error predicting positives (False Positives)

$$1 - \textit{Specificity} = 1 - \frac{TN_A}{TN_A + FP_A} = \frac{FP_A}{N_A}$$

Linear (LDA) & Quadratic (QDA) Discriminant Analysis

[FYI Only]

Bayes Theorem

[FYI Only]

Prior, Posterior and Classification

3 important Bayesian statistics, relevant to LDA (and QDA):

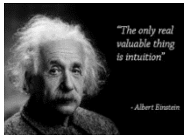
- Prior Probability: The proportion of 0's and 1's in the outcome variable → If we had no predictors (i.e., null model), this would be our only reference *(e.g., if on average 5% of patients are diagnosed with heart disease, then in the absence of predictors, your prior probability of having heart disease is 0.05 or 5%).*
- Posterior Probability: The probability of being classified as a 0 or a 1, given the value of predictors (e.g., the probability of having heart disease, given that a person smokes and is over 60 years of age is 0.2 or 20%, this figure is illustrative only)
- LDA (and QDA) computes a linear (or quadratic) discriminant function aimed at providing these posterior probabilities, which are much better predictors than prior probabilities.
- Classification: is our prediction of whether an observation will be a 0 or a 1, based on the posterior probability. Based on a given threshhold λ (e.g., prob > 0.5, prob > 0.6, etc.)

Bayes Theorem

Illustration

GPA		1	2	3	4	5	6	7	8	9	10
> = 3.5	1	A	A	A	A	A	A	NA	NA	NA	NA
	2	A	A	A	A	A	A	NA	NA	NA	NA
< 3.5	3	A	A	NA	NA	NA	NA	NA	NA	NA	NA
	4	A	A	NA	NA	NA	NA	NA	NA	NA	NA
	5	A	A	NA	NA	NA	NA	NA	NA	NA	NA
	6	A	A	NA	NA	NA	NA	NA	NA	NA	NA
	7	A	A	NA	NA	NA	NA	NA	NA	NA	NA
	8	A	A	NA	NA	NA	NA	NA	NA	NA	NA
	9	A	A	NA	NA	NA	NA	NA	NA	NA	NA
	10	A	A	NA	NA	NA	NA	NA	NA	NA	NA

P(A) = 28 / 100 =	0.28	Prior
P(A|GPA >= 3.5) = 12 / 20 =	0.60	Posterior
P(GPA >= 3.5) = 20 / 100 =	0.20	Prior
P(GPA >= 3.5|A) = 12 / 28 =	0.43	Posterior
P(A) * P(GPA >= 3.5|A) =	0.12	Bayes
P(GPA >= 3.5) * P(A|GPA >= 3.5) =	0.12	



Bayes Theorem: Intuition

- The Bayes Theorem is fundamental in statistics. A lot of what we do in predictive modeling is based on "Bayesian Statistics"
- The math behind Bayes Theorem is beyond the scope of this class, but it helps to understand intuitively what it means
- The Bayes Theorem describes the probability of an event based on given conditions (that may be related to the event).
- Non-Bayesian – e.g., (Prior) probability (make an A in this class)
  Bayesian – e.g., (Posterior) probability (make an A, given a GPA of 3.5)
- If GPA is related to grades, then the conditional probability based on GPA will be different than the general probability. In a nutshell, $P(A|B)$ and $P(B|A)$ are conditional probabilities:

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

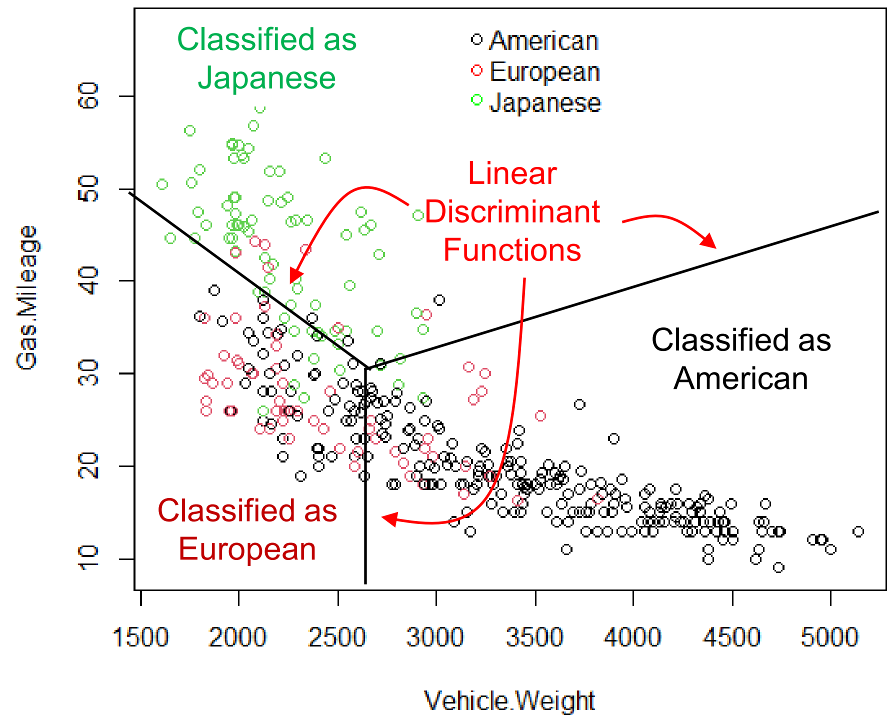
E.g., the probability of a student with a GPA>3.5 earning an A in the class is equal to the probability that a student who made an A has a GPA>=3.5, times the probability of anyone making an A, divided by the probability of anyone having a GPA>=3.5

Linear Discriminant Analysis (LDA)

[FYI Only]

LDA: Intuition

- Intuitively, LDA is a simple concept, and it can be best explained with a simple example model from the {ISLR}Auto data set, predicting the Origin of a vehicle, with K = 3 (American, European or Japanese) classes and p = 2 predictors Gas Mileage and Vehicle Weight
- We can plot the data for Gas Mileage vs. Vehicle Weight and color-code the dots by outcome category
- The idea behind LDA is to define 3 lines (one for each class) and rotate them until the classification accuracy is maximized
- These lines are called "Linear Discriminant Functions"
- It is easy to see visually that if the classes are easily separated LDA works well, but not so well if classes are comingled.



LDA Explained

- If we know the prior probability that $Y = k$ (an outcome is in class k) and we know the distribution of the predictors X, we can estimate the posterior probability that $Y = k$ for a given set of X predictors.
- Assuming a normal distribution of the X's, we can use Bayes Theorem to calculate the posterior probability $P(Y = k \mid X_1 = c_{1,} X_2 = c_{2,} \text{etc.})$
- LDA estimates K "Linear Discriminant Functions", one for each of the K possible categories and assigns the observation to the category k that has the highest posterior probability of being correctly classified
- If there are K outcome categories, there are K discriminant functions, one for each K
- Other types of discriminant analysis use non-linear discriminant functions (e.g., quadratic) of the X's.

LDA vs. Logistic Regression

- LDA accomplishes the same results as Logistic regression, but through a probabilistic method based on Bayes' Theorem
- LDA sometimes outperforms Logistic regression in terms of predictive accuracy, particularly when:
  1) The outcome is multinomial (> 2 categorical outcomes) :
  2) The classifications for each predictor are well separated
  3) The predictors are normally distributed
- The tradeoff is predictive accuracy vs. interpretability → Logistic regression is better for inference and interpretation
- Alternative modeling should be tested with cross-validation
- In sum, only use LDA if:
  - ✓ Interpretation and inference are less important
  - ✓ All predictor variables are continuous and normally distributed → no binary, categorical or discrete predictors
  - ✓ The model has better predictive accuracy than Logistic
  - ✓ Otherwise use Logistic Regression

LDA: Fit Statistics

- Like with other classification models, the confusion matrix needs to be evaluated to inspect:
  - Error Rates – proportion of incorrect classifications
  - Sensitivity – proportion of correct positive classifications
  - Specificity – proportion of correct negative classifications, or 1-Specificity – proportion of False Positives
- Depending on the analysis goals, we may want to place more emphasis on one or the other – is it better to send the innocent people to jail or to let the guilty go free?
- The Bayes classifier uses the threshold probability of 50% to classify an observation into a category, but this threshold can be changed depending on the analysis goals.

- Varying this threshold and plotting the resulting sensitivity and specificity values yields the ROC curve ("Receiver Operating Characteristics") – more on this later.

Linear Discriminant Analysis (LDA)

- {libary} Topic
  `Search keywords`

- Linear Discriminant Analysis (LDA)
  `library(ISLR)`

Tuning LDA

[FYI Only]

Confusion Matrix: Illustration

Example (see textbook): predicting loan defaults with LDA:

$$\text{Error Rate} = \frac{\text{Incorrect}}{\text{Total}} = \frac{23 + 252}{10{,}000} = 0.028$$

$$\text{Sensitivity} = \frac{\text{Correct Defaults}}{\text{Total Defaults}} = \frac{81}{333} = 0.243$$

$$\text{Specificity} = \frac{\text{Correct NoDefaults}}{\text{Total NoDefaults}} = \frac{9{,}644}{9{,}667} = 0.997$$

$$1 - \text{Specificity} = \text{False Positives} = 1 - 0.997 = 0.003$$

Predicted Default	Actual Default		Total
	No	Yes	
No	9,644	252	9,896
Yes	23	81	104
Total	9,667	333	10,000

- Interestingly, this model has a low error rate of 2.8%; but we know a priori that 3.3% (333/10,000) of the clients default, so we could predict that no one will default and be correct 96.7% of the time → The model is not a big help in this respect.
- Similarly, the models does a poor job (worse than flipping a coin) at predicting actual defaults
- In contrast, it does a very nice job at predicting no-defaults (i.e., low positive error)

Tuning LDA: The Threshold λ

- LDA uses a "classifier threshold" $\lambda = \text{Prob}(\text{Default} = \text{Yes}) > 0.5$ by default. So, only loans with more than a 50% chance of defaulting are classified as expected defaults. But what if we change this threshold to a more conservative 20%? See what happens:

$$\text{Error Rate} = \frac{\text{Incorrect}}{\text{Total}} = \frac{253 + 138}{10{,}000} = 0.039$$

$$\text{Sensitivity} = \frac{\text{Correct Defaults}}{\text{Total Defaults}} = \frac{195}{333} = 0.586$$

$$\text{Specificity} = \frac{\text{Correct NoDefaults}}{\text{Total NoDefaults}} = \frac{9{,}432}{9{,}667} = 0.976$$

$$1 - \text{Specificity} = \text{Positive Error} = 1 - 0.976 = 0.024$$

Predicted Default	Actual Default		Total
	No	Yes	
No	9,432	138	9,570
Yes	235	195	430
Total	9,667	333	10,000

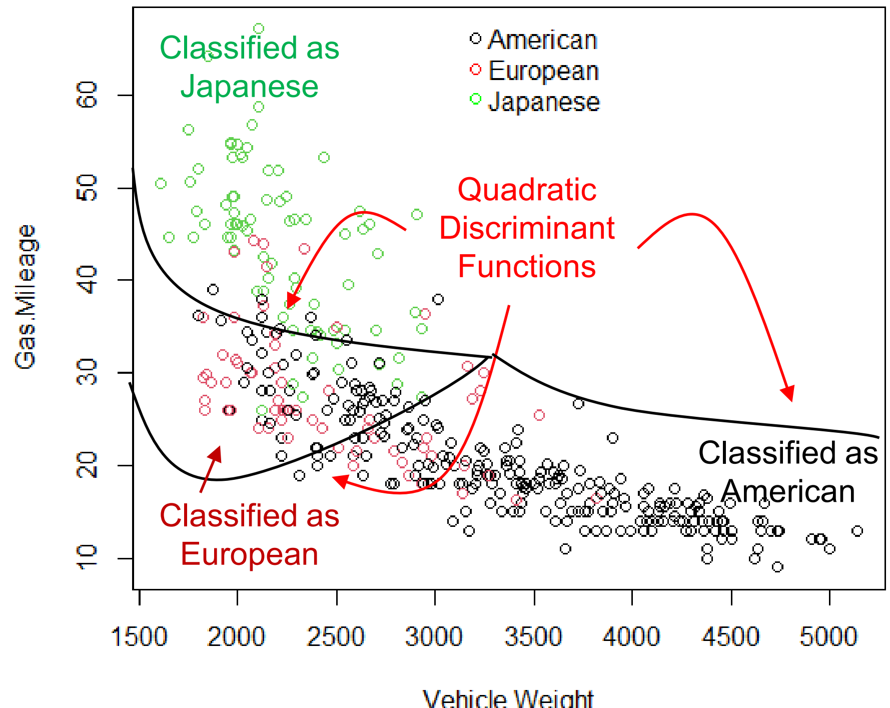
- Error Rate ↑; Sensitivity ↑; Specificity ↓
- Importantly, the prediction of defaults has improved from 24.3% to 58.6% (better than flipping a coin)
- Again, tuning the model to our analysis goals is key!! If your business goal is to predict defaults, $\lambda = 0.2$ is better than $\lambda = 0.5$

Quadratic Discriminant Analysis (QDA)

[FYI Only]

QDA: Intuition

- QDA is just like LDA, but the discriminant functions are quadratic
- We follow the earlier LDA example predicting the Origin of a vehicle, with K = 3 (American, European or Japanese) classes and p = 2 predictors Gas Mileage and Vehicle Weight
- The idea behind QDA is to define 3 quadratic curves (one for each class) and move them until the classification accuracy is maximized
- The curves are called "Quadratic Discriminant Functions"
- Which to use, LDA or QDA?
  - Whichever discriminates the outcome in the data, yielding fewer misclassifications
  - As with polynomial regression models QDA has better training fit, more over-fitting problems, and more variance than LDA, but it is a more flexible classifier



Quadratic Discriminant Analysis (QDA)

- {libary} Topic
  `Search keywords`
- Quadratic Discriminant Analysis (QDA)
  `library(MASS)`

Which Classification Model to Use?

Example: Comparing all Models

```
{r}
round(rbind(logit.rates.50, logit.rates.60,
            lda.rates.50, lda.rates.60,
            qda.rates.50, qda.rates.60),
      digits = 3)
```

	Accuracy	Error	Sensitivity	Specificity	False Pos
logit.rates.50	0.698	0.302	0.469	0.822	0.178
logit.rates.60	0.676	0.324	0.327	0.867	0.133
lda.rates.50	0.683	0.317	0.408	0.833	0.167
lda.rates.60	0.676	0.324	0.347	0.856	0.144
qda.rates.50	0.655	0.345	0.429	0.778	0.222
qda.rates.60	0.676	0.324	0.408	0.822	0.178